Siddeshwar "Sid" Raghavan

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EDUCATION

Purdue University — West Lafayette, Indiana, USA
Ph.D. Electrical and Computer Engineering — *Jan 2022 – Present*
Advisors: Prof. Fengqing Zhu and Prof. Edward Delp

University of Wisconsin-Madison — Madison, Wisconsin, USA
M.S Electrical Engineering (Research) — *Aug. 2019 – May 2021*
Advisors: Prof. Yin Li and Prof. Andreas Velten

PSG College of Technology — Coimbatore, India
B.E Electronics and Communication Engineering — *Jul. 2014 - May 2018*

RESEARCH EXPERIENCE

Graduate Research Assistant - Advised by Dr. Fengqing Zhu, Dr. Edward Delp — May 2022 – Present
VIPER Lab, Purdue University

- **Continual Learning in Computer Vision:** Developed online and long-tailed continual learning methods for food image classification, distribution shift adaptation, and exemplar-free learning with vision-language alignment strategies **[WACV 2024, CVPRW 2024, AAAI 2026]**. Developed ATLAS, a continual audio-visual segmentation framework for non-stationary streams, introducing parameter-stabilized LoRA for stable parameter-efficient adaptation **[Under Review]**
- **End-to-end ML Optimization for Dietary Monitoring:** Co-developed EgoCal, an egocentric calorie-tracking algorithm using live streams from Meta Aria glasses, vision-language alignment, and Kalman-inspired adaptive temporal fusion **[CVPRW 2026]** (★ Best Paper Runner-Up, provisional patent). Developed vision-language methods for dietary assessment from before-and-after images and stereo-image volume perception for nutrition estimation **[2x CVPRW 2026]**.
- **Implicit Representations and 3D Vision:** Co-developed real-scale 3D reconstruction algorithms for food portion estimation from monocular images through vision-language understanding **[IEEE CAI 2026]**. Mentored a 3D portion estimation project reconstructing point clouds from 2D food images for portion estimation **[ICPR 2024]**.
- **Embodied AI & Super-Resolution (Ongoing):** Developing unsupervised continual skill discovery algorithm for robotic imitation learning, and designing gated video super-resolution algorithms for enhancing sparse and noisy data.

Graduate Research Assistant - Advised by **Dr. Yin Li, Dr. Andreas Velten** — Jan 2020 – May 2021
University of Wisconsin-Madison

- Developed a computer vision system for reconstructing intensity images from **Non-Line-of-Sight (NLOS)** measurements by combining implicit representations with supervised learning from 2D scene images. **[IEEE TPAMI 2022]**
- Final thesis - **2D NLOS Human Pose Estimation** using a hybrid CNN and LSTM network.

TEACHING EXPERIENCE

Purdue University — Aug 2024 – May 2026
VIP Projects

- **Instructor of Record**, Spring 2026; **Graduate Teaching Assistant**, Fall 2024–Fall 2025.
- Designed project objectives, milestones, and assessment framework; mentored undergraduate honors students and supported course infrastructure for code and deliverables.
- Received the **VIP Instructor of Record Award in Spring 2026**.

University of Wisconsin–Madison — Jan 2021 – May 2021
Lead Graduate Teaching Assistant – Digital Fundamentals

- Designed a semi-supervised ModelSim grading framework and received a Teaching Excellence Award for for fostering student engagement and community in a virtual environment.

INTERNSHIPS

Givens Associate - Advised by Dr. Tanwi Mallick May 2025 - July 2025
Argonne National Laboratory, USA

- Selected as **one of thirteen** Givens Associates this year out of 400 interns
- Developed a multi-agentic LLM framework in a post-training setup for scientific coding and reasoning. Surpassed current works by 10%.**[NeurIPS DL4C 2025][Paper]**

Engineering Intern Sept 2018– May 2019
Adori Labs, India

- Identified emerging markets for audio engineering and engagement and rapidly prototyped solutions for the domains. I developed a wake up voice SDK for the Adori player, Amazon Alexa and Google Home compatible frameworks and an in-house optmized search module with Elasticsearch. (**Skills: Python, Swift, C++, AWS/GCloud, REST APIs, User-study, UX design**)

Research Intern - advised by Dr. Rajbabu Velmurugan Jun 2017 – Jul 2017
Indian Institute of Technology Bombay, India

- I worked on a weakly supervised **Visual SLAM** project and developed a single camera system (rather than a conventional multi-camera setup)

Research Intern - advised by Dr. Ashok Jhunjhunwala Jun 2016 – Jul 2016
Indian Institute of Technology Madras, India

- Designed and built a real-time battery state of health monitor for lithium ion cells with TI low power boards

RESEARCH PUBLICATIONS AND PATENTS

1. Can You Hear, Localize, and Segment Continually? An Exemplar-Free Continual Learning Benchmark for Audio-Visual Segmentation (**Siddeshwar Raghavan**, Gautham Vinod, Bruce Coburn, Fengqing Zhu) **[ArXiV 2026][Paper] [Code]**

2. PANDA - Patch And Distribution-Aware Augmentation for Long-Tailed Exemplar-Free Continual Learning (**Siddeshwar Raghavan**, Jiangpeng He, Fengqing Zhu) **[AAAI 2026][Paper][Code]**

3. Visual-Temporal Fusion For Continuous Object Weight Estimation From Video **[US Provisional Patent]** (Fengqing Maggie Zhu, Bruce Barbet Coburn, Siddeshwar Raghavan, Gauthan Vinod)

4. EgoCal: Combining Egocentric Data with Vision-Language Fusion for Real-time Dietary Tracking (Gautham Vinod*, **Siddeshwar Raghavan***, Bruce Coburn*, Wei-Lun Chu, Jinge Ma, Fengqing Zhu) **[CVPRW 2026] [Paper] [Code]**(* Equal contribution) *(★ Best Paper Runner Up)*

5. DietDelta: A Vision-Language Approach for Dietary Assessment via Before-and-After Images (Gautham Vinod, **Siddeshwar Raghavan**, Bruce Coburn, Fengqing Zhu) **[CVPRW 2026] [Paper] [Code]** *(★ Best Paper Nominee)*

6. Not Your Stereo-Typical Estimator: Combining Vision and Language for Volume Perception (Gautham Vinod*, Bruce Coburn*, **Siddeshwar Raghavan***, Fengqing Zhu) **[CVPRW 2026] [Paper] [Code]**

7. Implicit-Scale 3D Reconstruction for Multi-Food Volume Estimation from Monocular Images (Yuhao Chen, Gautham Vinod, **Siddeshwar Raghavan**, Talha Ibn Mahmud, Bruce Coburn, Jinge Ma, Fengqing Zhu, Jiangpeng He) **[ArXiV 2026] [Paper]**

8. MOSAIC: Multi-agent Orchestration for Task-Intelligent Scientific Coding (**Siddeshwar Raghavan**, Tanwi Mallick) **[NeurIPS DL4C 2025][Paper]**

9. DELTA: Decoupling Long-Tailed Online Continual Learning (**Siddeshwar Raghavan**, Jiangpeng He, Fengqing Zhu) **[CVPR Workshop 2024][Paper][Code]**

10. Online Class-Incremental Learning for Real-World Food Image Classification (**Siddeshwar Raghavan**, Jiangpeng He, Fengqing Zhu) **[WACV 2024][Paper][Code]**

11. Size Matters: Reconstructing Real-Scale 3D Models from Monocular Images for Food Portion Estimation (Gautham Vinod*, Bruce Coburn*, **Siddeshwar Raghavan***, Jiangpeng He, Fengqing Zhu) **[IEEE CAI 2026]** [Paper] [Code]
12. MetaFood3D: Large 3D Food Object Dataset with Nutrition Values (Yuhao Chen†, Jiangpeng He†, Gautham Vinod*,**Siddeshwar Raghavan***, Chris Czarnecki, Jinge Ma, Talha Ibn Mahmud, Dayou Mao, Saeejith Nair, Pengcheng Xi, Alexander Wong, Edward Delp, Fengqing Zhu) **[ArXiV 2024]** [Paper] [Website] (*, † Equal contribution)
13. MFP3D: Monocular Food Portion Estimation Leveraging 3D Point Clouds (Jinge Ma, Xiaoyan Zhang, Gautham Vinod, **Siddeshwar Raghavan**, Jiangpeng He and Fengqing Zhu) **[MADiMa 2024]** [Paper]
14. Physics to the rescue: Deep non-line-of-sight reconstruction for high-speed imaging (Fangzhou Mu, Sicheng Mo, Jiayong Peng, Xiaochun Liu, Ji Hyun Nam, **Siddeshwar Raghavan**, Andreas Velten, Yin Li) **[IEEE TPAMI 2022]** [Paper] [Code]

AWARDS AND RECOGNITION

- Received the **VIP Instructor of Record Award** in recognition of outstanding leadership, inclusive mentorship, and guidance of undergraduate research teams (Spring 2026)
- Best Paper Runner up - EgoCal: Combining Egocentric Data with Vision-Language Fusion for Real-time Dietary Tracking (CVPRW 2026)
- Teaching Excellence during COVID (UW–Madison, Spring 2021) - Recognized for cultivating virtual student communities and enhancing engagement as Head TA during remote instruction.

PROFESSIONAL SERVICE AND LEADERSHIP

Professional Service

- **Workshop Organizer – MetaFood, CVPR 2025**
  Main organizer of an international workshop featuring two challenges: (1) 3D food reconstruction from monocular images, and (2) vision–language models for interpreting textual food descriptions.
- **Reviewer – Premier Machine Learning and Computer Vision Venues**
  Served as reviewer for top conferences and journals, contributing to the scholarly review process. [NeurIPS-26, ICPR-26, ICLR-26, AAAI-26, WACV-25, AAAI-25, NeurIPS-25, ICPR-24, MMSP-24, Elsevier, ICIP-24, CVPRW-24, ICME-23, CVPR-23, NeurIPS-23]
- **Judge – Undergraduate Research Conference, Purdue (2025)**
  Lead graduate judge evaluating research posters and oral presentations across diverse disciplines.

Leadership and Outreach

- **Graduate Student Mentor and Teaching Leadership (VIP, Purdue University)**
  Led project frameworks in signal processing and computer vision, mentored undergraduate honors students, and served as Head TA mentoring other teaching assistants.
- **Workshop Instructor – Deep Learning (2025, Purdue University)**
  Conducted a 2-hour professional development workshop for 50+ undergraduate honors students, introducing fundamentals of deep learning.
- **Purdue Graduate Student Government – Life Team Co-Chair**
  Managed funds and organized large-scale social and well-being events serving 10,000+ graduate students.

MENTORING EXPERIENCE

Undergraduate Students

- Joon Kim — Accepted into the M.S. ECE program, Purdue University
- Piotr Nabrzyski — Awarded Summer Research Fellowship, University of Houston
- Arnav Singh - Mentored in VIP, accepted at IBM for summer intern 2026
- Donny Weintz - Mentored in VIP, Won leadership award Spring 2026 and accepted at GE Aerospace May 2026

Graduate Students

- Jinge Ma — Co-authored paper accepted at MADiMa 2024
- Gautham Vinod — 1 Co-authored paper accepted at IEEE CAI 2026, 3 co-authored papers at CVPRW 2026

REFERENCES

Prof. Fengqing Zhu	Purdue University	zhu0@purdue.edu
Prof. Edward Delp	Purdue University	ace@purdue.edu
Prof. Xiaoqian Wang	Purdue University	joywang@purdue.edu
Prof. Andreas Velten	University of Wisconsin-Madison	velten@wisc.edu
Prof. Yin Li	University of Wisconsin-Madison	yin.li@wisc.edu